

Advanced Programs in Project

CS 411

Project Track 1

Transactions:

- 1) `<h2>General Best Pharmacies</h2> <button
onClick={fetchBestPharmacies}>Load Best Pharmacies</button>
{bestPharmaciesError ? (<p>There was an error loading the best pharmacies.
Please try again</p>) : bestPharmaciesLoading ? (<p>Loading...</p>) : (
{bestPharmacies.map((pharmacy, index) => (<li key={index}>
{pharmacy.address}, {pharmacy.city}, {pharmacy.state}))})}`
- 2) `const fetchBestPharmacies = async () => { setBestPharmaciesLoading(true);
setBestPharmaciesError(false); try { const response = await fetch(
http://localhost:4000/api/getUserMedsAndPharmacies?userEmail=${encodeURIComponent(
userEmail || ""
)}); if (!response.ok) { setBestPharmaciesError(true); throw new
Error(HTTP error! status: ${response.status}); } const data = await
response.json(); console.log("Best pharmacies data:", data); // Extract
pharmacies from the response const pharmaciesData = data.pharmacies || [];
interface PharmacyRaw { address: string; city: string; state: string; } const
data_transformed = pharmaciesData.map((pharmacy: PharmacyRaw) => ({
address: pharmacy.address, city: pharmacy.city, state: pharmacy.state, }));
setBestPharmacies(data_transformed); } catch (err) { console.error("Error
fetching best pharmacies:", err); setBestPharmaciesError(true); } finally {
setBestPharmaciesLoading(false); } };`

Triggers:

- 1) `CREATE TRIGGER `limit_currentlyHas` BEFORE
INSERT
ON
`stage2_schema`.`currentlyHas` FOR EACH ROW
BEGIN
DECLARE
cnt INT;
SELECT
COUNT(*) INTO cnt
FROM
`stage2_schema`.`currentlyHas`
WHERE
user_email = NEW.user_email;`

```

IF
    cnt >= 5 THEN SIGNAL SQLSTATE '45000'
SET
    MESSAGE_TEXT = 'Error: A user cannot have more than 5 currentlyHas records.';
END IF
;
END

```

PROCEDURE:

```

1) CREATE PROCEDURE
    stage2_schema.getMedicinesBySymptomsAdvanced(IN user_email_param
VARCHAR(255)) READS SQL DATA
BEGIN
    CREATE TEMPORARY TABLE tempUserSymptoms AS
    SELECT `symptom_name`
    FROM `stage2_schema`.`currentlyHas`
    WHERE `user_email` = user_email_param;

    SELECT
        t.`symptom_name`,
        COUNT(DISTINCT ch.`user_email`) AS user_count
    FROM
        tempUserSymptoms t
    JOIN
        `stage2_schema`.`currentlyHas` ch
    ON
        t.`symptom_name` = ch.`symptom_name`
    GROUP BY
        t.`symptom_name`
    ORDER BY
        user_count DESC;

    SELECT
        m.`name` AS medicine_name,
        m.`rating` AS medicine_rating,
        COUNT(tb.`symptom_name`) AS symptoms_treated
    FROM
        `stage2_schema`.`medicines` m
    JOIN
        `stage2_schema`.`treatedBy` tb
    ON
        m.`name` = tb.`medicine_name`
    JOIN
        tempUserSymptoms t
    ON
        tb.`symptom_name` = t.`symptom_name`
    GROUP BY

```

```
    m.`name`,  
    m.`rating`  
ORDER BY  
    symptoms_treated DESC,  
    medicine_rating DESC  
LIMIT  
    15;  
DROP  
    TEMPORARY TABLE tempUserSymptoms;  
END
```